

CSCE 670 INFORMATION RETRIEVAL · MODULE 10 · LECTURE NOTES

Web-Scale Retrieval

What changes when the collection is adversarial.

MODULE OUTCOMES

- Explain crawling and its politeness constraints.
- Explain duplicate detection at web scale.
- Explain link analysis and what it measures.
- Explain adversarial retrieval and spam.
- Explain the architecture of a web search system.

1 Crawling

THE MECHANICS

a frontier of URLs to fetch, prioritised
politeness: a delay per host, robots.txt
respected, and a contactable user agent
string
duplicate URL detection before fetching
and a recrawl policy driven by observed change
rate

THE PRIORITISATION PROBLEM

you cannot fetch everything, so which pages
first? Discovery order, link-based importance,
and observed update frequency all contribute, and
the policy determines what your index contains.

THE TRAPS

infinite URL spaces -- calendars, session ids,
faceted navigation with combinatorial
filters. A crawler without a depth and
pattern limit will not escape them.
and soft 404s, which return status 200 with an
error page, so the crawler happily keeps the
garbage and indexes it

AND POLITENESS IS NOT OPTIONAL.

A crawler without a per-host rate limit is a denial of service, whatever was intended — which is CSCE 701 Module 01 §4's authorisation constraint arriving in this context, and it is a professional and in some jurisdictions legal matter rather than a courtesy. **The politeness-as-obligation framing belongs**

here rather than in a footnote: a student crawler that takes down a small site has caused real harm, and the mechanism is a missing three-line delay.

2 Duplicates

A LARGE FRACTION OF THE WEB IS DUPLICATE OR NEAR-DUPLICATE, AND INDEXING IT WASTES EVERYTHING

Mirrors, syndicated news content, printer-friendly variants, session-id URLs pointing at identical pages, and templated pages differing in one field — a very large share of fetched pages are substantially duplicates of others, by every published measurement.

Which costs on every axis at once: crawl bandwidth spent fetching them, index space storing them, and result quality when they are returned — since a result page containing ten near-identical documents is worth approximately one (Module 05 §3's redundancy problem).

So shingling, minhashing, and locality-sensitive hashing (CSCE 676 Module 02) are deployed exactly here — and this is their original application, which is why that module's worked examples are documents rather than something more abstract.

And canonicalisation is the other half of the problem: URL normalisation, and then choosing one canonical version from each duplicate set — which is a policy decision with real consequences for whoever is not chosen (Module 12's exposure argument), and which site owners care about a great deal.

3 Links

A LINK IS AN ENDORSEMENT, WHICH MAKES THE LINK GRAPH EVIDENCE THE TEXT DOES NOT CONTAIN

A page's own text is written by its author, and its inbound links are written by other people — which makes links the first large-scale source of retrieval evidence not under the document's own control, and that independence was the whole insight.

PageRank (CSCE 676 Module 07 §2) formalises this recursively: an endorsement from a well-endorsed page counts for more than one from an obscure page, which is computed as a stationary distribution.

And anchor text is at least as valuable as the link itself: it is how other people describe a page, which addresses Module 01 §2's vocabulary mismatch directly — because the describers use the searcher's words rather than the author's, which is exactly the asymmetry that module identified.

But the link graph became adversarial, which is §4 — and the general result is worth stating: once a signal is known to matter, it is manufactured.

- **As one feature among hundreds in a learned ranker (Module 06 §3) — rather than as the ranking function, which it never quite was even in the systems most associated with it.**
- **Anchor text remains valuable, and is harder to manufacture convincingly at scale than a link is, because it requires producing plausible varied text on pages that look legitimate.**

- **And link-based spam detection** uses the graph's *structure* rather than its scores — **unnatural link patterns are detectable as patterns** even when the individual links look fine, which is CSCE 676 Module 07's community detection put to a defensive use.
- **While the raw score's value fell** substantially as manipulation techniques improved — **which is a general lesson about any published ranking signal** rather than a fact about links.
- **So: use it, weight it alongside everything else, and expect it to be attacked. Once a signal is known to matter, it is manufactured** — which means **a ranking signal's value decays after it becomes public**, and **is an argument for many weak signals over one strong one** (§4, and CSCE 701 Module 01's defence in depth).

4 Adversarial retrieval

ON THE WEB, THE DOCUMENTS ARE WRITTEN BY PEOPLE WHO WANT TO RANK, WHICH CHANGES EVERYTHING

In an ordinary document collection the documents are entirely indifferent to your ranking function. On the web they are written specifically to influence it — which makes retrieval an adversarial problem rather than a statistical one, and that is the single structural difference from everything else in this course.

So every signal you use will be attacked: keyword stuffing, hidden text, cloaking (serving different content to the crawler), link farms, purchased links, and now generated content at scale — each of which appeared within a short time of the corresponding signal becoming important.

And the defences are the ones from CSCE 701: many weak signals rather than one strong one, detection rather than prevention (Module 09 of that course), and a preference for signals that are expensive to fake — user behaviour over time, reputation, and editorial judgement.

Which makes web search a security problem with a ranking component rather than the reverse — and it is why the ranking signals are not published, which is a defensive decision rather than a purely commercial one, and is one of the few legitimate uses of obscurity (CSCE 701 Module 01 §1).

- A crawler and a frontier (§1), with duplicate detection and canonicalisation (§2) in front of the index.
- A document-partitioned index across many machines (Module 04 §4), usually tiered by document quality so that most queries are answered from a small high-quality tier (Module 04 §3).
- A cheap first-stage retrieval, then a learned ranker (Module 06 §4), then presentation logic that decides what kind of result page to build (Module 09 §4's intent).
- Query understanding in front of all of it (Module 09), and caching at every layer (Module 04 §4).
- And a spam and quality pipeline running continuously beside the whole thing — which is as large as the retrieval system. The quality and anti-abuse pipeline being comparable in size to the retrieval system is the fact most surprising to people who have only built retrieval, and it follows directly from §4's callout.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Manning, Raghavan & Schütze, chapters 19 through 21 (free)	§1 through §3 — crawling, duplicate detection, and link analysis, with the algorithms.
Croft, Metzler & Strohman, chapters 3 and 9 (free PDF)	§1's mechanics and §4's spam material, in more practical detail.

Resource	What it covers for this module
Brin & Page — The Anatomy of a Large-Scale Hypertextual Web Search Engine (free)	§3 and §4 in the original — and worth reading for how much of the architecture survived.
Gyöngyi & Garcia-Molina — Web Spam Taxonomy (free)	§4's techniques, catalogued — the defensive equivalent of CSCE 701's threat modelling.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Write a politely rate-limited crawler** for a site you own, and verify the delay.
- 2 **Read a real robots.txt** and implement its rules.
- 3 **Find an infinite URL space** on a site you own and construct the pattern limit that escapes it.
- 4 **Detect soft 404s** by content rather than status code.
- 5 **Run minhash near-duplicate detection** over your crawl and report the duplicate rate.
- 6 **Choose canonical versions** and say what rule you used.
- 7 **Compute PageRank** on a crawled subgraph.
- 8 **Extract anchor text** and compare it against the target pages' own titles.
- 9 **Find an unnatural link pattern** in a graph you have.
- 10 **List the signals in your own system** and say how each could be manufactured.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Give the crawling mechanics and the two traps.
- 2 Why is politeness an obligation rather than a courtesy?
- 3 Why does web duplication cost on every axis?
- 4 Which CSCE 676 machinery applies, and why is this its original application?
- 5 Why is a link evidence the text does not contain?
- 6 Why is anchor text particularly valuable?
- 7 State the general result about published signals.
- 8 What is link analysis worth now, and in what role?
- 9 What makes web retrieval adversarial, and what are the defences?

10 Name six components of a web search system.